(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
06 April 2023 (06.04.2023)

WIPO | PCT

(10) International Publication Number
WO 2023/052652 A1

(51) International Patent Classification:
A61K 31/415 (2006.01) *A61P 9/00* (2006.01)
C07D 231/12 (2006.01) *A61P 25/02* (2006.01)

(21) International Application Number:
PCT/EP2022/077587

(22) International Filing Date:
04 October 2022 (04.10.2022)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:
21200394.1 01 October 2021 (01.10.2021) EP

(71) Applicant: **SANDOZ AG** [CH/CH]; Lichtstr. 35, 4056 Basel (CH).

(72) Inventors: **ADAMER, Verena**; c/o Sandoz GmbH, Biochemiestr. 10, 6250 Kundl (AT). **PICHLER, Arthur**; c/o Sandoz GmbH, Biochemiestr. 10, 6250 Kundl (AT). **MARGREITER, Renate**; c/o Sandoz GmbH, Biochemiestr. 10, 6250 Kundl (AT).

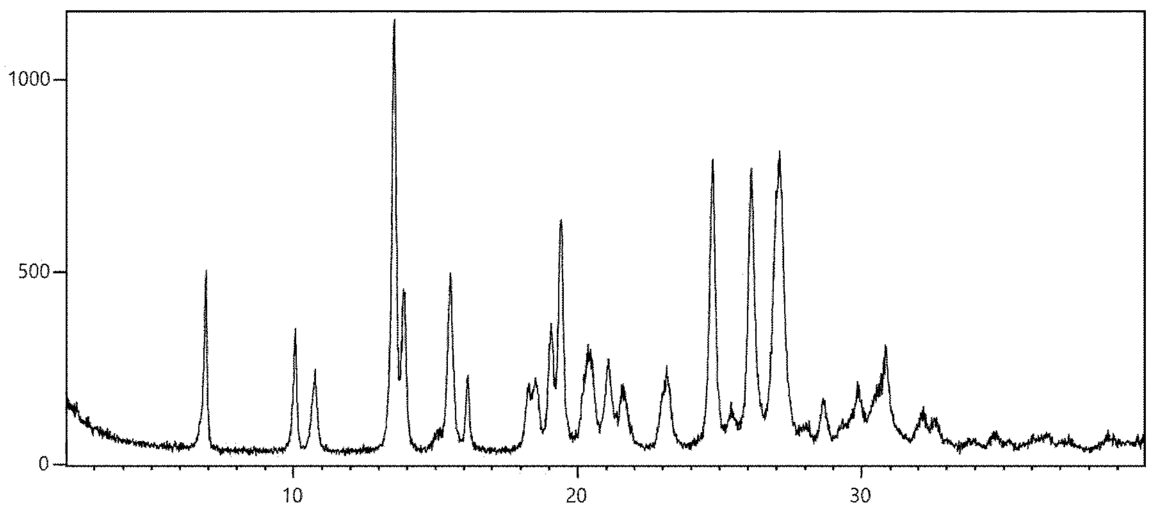
(74) Agent: **PRÜFER & PARTNER MBB**; Sohnckestr. 12, 81479 München (DE).

(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CV, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, ME, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK,

(54) Title: CRYSTALLINE FORM OF ACORAMIDIS HYDROCHLORIDE

Figure 1



(57) Abstract: The present invention relates to a crystalline form of acoramidis hydrochloride and a process for its preparation. Furthermore, the invention relates to a pharmaceutical composition comprising the crystalline form of acoramidis hydrochloride of the present invention and at least one pharmaceutically acceptable excipient. The pharmaceutical composition of the present invention can be used as a medicament, in particular for the treatment of TTR amyloidosis (ATTR).

WO 2023/052652 A1

[Continued on next page]

CRYSTALLINE FORM OF ACORAMIDIS HYDROCHLORIDE

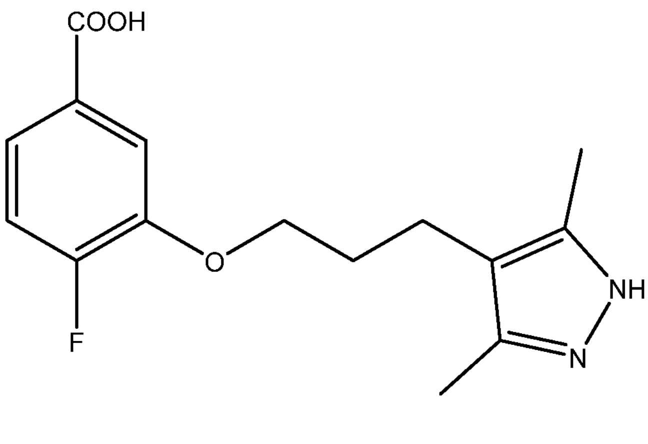
FIELD OF THE INVENTION

The present invention relates to a crystalline form of acoramidis hydrochloride and a process for its preparation. Furthermore, the invention relates to a pharmaceutical composition comprising the crystalline form of acoramidis hydrochloride of the present invention and at least one pharmaceutically acceptable excipient. The pharmaceutical composition of the present invention can be used as a medicament, in particular for the treatment of TTR amyloidosis (ATTR).

BACKGROUND OF THE INVENTION

Acoramidis, formerly AG10, is a potent and selective transthyretin (TTR) stabilizer being developed to treat TTR amyloidosis (ATTR). ATTR is a progressive, fatal disease in which deposition of amyloid derived from either mutant or wild type TTR causes severe organ damage and dysfunction. Clinically, ATTR presents predominantly as either TTR amyloid cardiomyopathy (ATTR-CM) or as peripheral polyneuropathy (ATTR-PN) [Jonathan C.Fox et al. First in-Human-Study of AG10, a Novel, Oral, Specific, Selective, and Potent Transthyretin Stabilizer for the Treatment of Transthyretin Amyloidosis: A Phase I Safety, Tolerability, Pharmacokinetic, and Pharmacodynamic Study in Healthy Adult Volunteers, Clinical Pharmacology in Drug Development 2019, 00(0) 1-15].

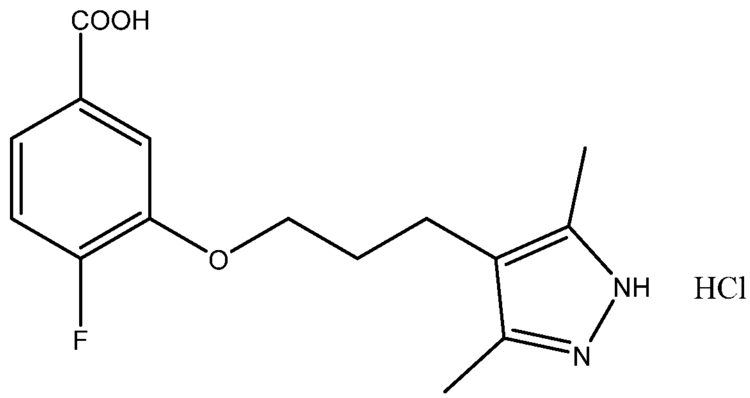
The chemical name of acoramidis is 3-(3-(3,5-dimethyl-1*H*-pyrazol-4-yl)propoxy)-4-fluorobenzoic acid. Acoramidis can be represented by the following chemical structure according to Formula (A)



Formula (A).

CLAIMS

1) A crystalline form of acoramidis hydrochloride (Form 1) according to the chemical structure as depicted in Formula (B)



Formula (B).

characterized by having a powder X-ray diffractogram comprising reflections at 2-Theta angles of (6.9 ± 0.2)°, (13.6 ± 0.2)° and (19.5 ± 0.2)°, when measured at a temperature in the range of from 20 to 30°C with Cu-Kalpha$_{1,2}$ radiation having a wavelength of 0.15419 nm.

2) The crystalline form of claim 1 characterized by having a powder X-ray diffractogram comprising additional reflections at 2-Theta angles of (10.1 ± 0.2)° and/or (10.8 ± 0.2)°, when measured at a temperature in the range of from 20 to 30°C with Cu-Kalpha$_{1,2}$ radiation having a wavelength of 0.15419 nm.

3) Use of the crystalline form as defined in claims 1 or 2 for the preparation of a pharmaceutical composition.

4) A pharmaceutical composition comprising a predetermined and/or effective amount of the crystalline form as defined in any one of claims 1 or 2 and at least one pharmaceutically acceptable excipient.

5) The pharmaceutical composition according to claim 4, which is an oral solid dosage form.

6) The pharmaceutical composition of claim 5, wherein the oral dosage form is a capsule or a tablet.

7) The crystalline form as defined in claims 1 or 2 or the pharmaceutical composition as defined in any one of claims 4 to 6 for use as a medicament.

Figure 1